Math 115A Final

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- Write your full name and student ID number clearly in the spots above.
- This exam contains 8 pages (including this cover page) and 6 questions for a point total of 80.
- The exam is 3 hours. The only outside resource allowed on the exam is your handwritten 4×6 note card.
- Write your graded work and solutions to each question on the front of each page. Draw a box around your final answer. You can use the back of each page for scratch work.

Question	Points	Score
1	10	
2	10	
3	10	
4	20	
5	10	
6	20	
Total:	80	

1. (10 points) Determine whether the statement is true or false. For true statements, provide justification. For false statements, find a counterexample. Each statement is worth 2 points.

(a) The real vector spaces $\mathbb{R}^{n^2}$ and $M_{n\times n}(\mathbb{R})$ are isomorphic.

True

$\mathbb{R}^{n^2}$ has dimension n^2

$M_{n\times n}(\mathbb{R})$ has dimension n^2

(b) Let T and U be linear operators on a vector space V. If $U \circ T$ is the zero transformation, then $T \circ U$ is the zero transformation.

$T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $U = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ False

$U \circ T = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$T \circ U = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$

If ~~T((a,b),(c,d)) = ((a,b),(0,0))~~

~~U((a,b),(c,d)) = ((0,b),(0,0))~~

~~T((a,b)) = (a+b,0)~~ $T(a,b) = (a+b, 0)$

~~U((a,b)) =~~ $U(a,b) = (b, b)$

then $U \circ T(a,b) = U(a+b, 0) = (0,0)$

but $T \circ U(a,b) = T(b,b) = T(2b)$

(c) If $T : V \to W$ is linear, then it sends linearly independent sets in V to linearly independent sets in W.

~~True.~~

~~Let $\{v_1, \dots, v_n\}$ be a linearly independent set in V.~~

~~Then $a_1v_1 + \dots + a_nv_n = 0$ for only $a_1 = \dots = a_n = 0$.~~

~~So, $T(a_1v_1 + \dots + a_nv_n) = 0$ since T is linear~~

~~$T(a_1v_1) + \dots + T(a_nv_n) = 0$~~

~~$a_1T(v_1) + \dots + a_nT(v_n) = 0$~~

False. Let $T = T_0$ ~~then~~ in $\mathbb{R}^2$. Then

$\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\} \to \left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$

in V in W

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(d) Let T be a linear operator on a finite dimensional vector space V. Then for any ordered bases $\mathcal{B}$ and $\mathcal{C}$, the matrices $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{C}}^{\mathcal{C}}$ are similar.

True

~~Let $\dim(V)$, $\beta = (b_1, \ldots, b_n)$, $C = (c_1, \ldots, c_n)$~~

Since β, C are ordered basis, $\exists$ $[T]_\beta^C$ and $[T]_C^\beta$

Thus, $[T]_\beta^\beta = [T]_C^\beta [T]_C^C [T]_\beta^C$

$= [T]_\beta^{C^{-1}} [T]_C^C [T]_\beta^C$

So $[T]_\beta^\beta$ and $[T]_C^C$ are similar

(e) Let T be a linear operator on an n-dimensional vector space V. If T is diagonalizable, then T has n distinct eigenvalues.

False let $T(x,y) = (x,y)$ on $\mathbb{R}^2$

Then $[T]_{std}^{std} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

and $\det([T]_{std}^{std} - \lambda I) = \begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 = 0$ so $\lambda = 1$

But $[T]_{std}^{std}$ is ~~already~~ diagonalizable since it is already diagonal but it only has 1 distinct eigenvalue

/ 0

2. (10 points) Let $T: V \to V$ be a linear operator. Prove that $\ker(T) \subset \ker(T^2)$.

Let $v \in \ker(T)$. Then, by definition, $T(v)=0$.

Now, we know that $T(T(v)) = T(0) = 0$ (apply T on both sides) because T is linear, so we know that $T(T(v))=0$, so $T^2(v)=0$

$v \in \ker(T^2)$.

Since any $v \in \ker(T)$ is also $v \in \ker(T^2)$, $\ker(T) \subset \ker(T^2)$.

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3. (10 points) Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by $T(a_1, a_2) = (2a_1 + a_2, a_2, a_1 - a_2)$. Let $\mathcal{B}$ be the standard ordered basis for $\mathbb{R}^2$. Let $\mathcal{C} = ((1, 0, 1), (1, 0, 0), (0, 1, -1))$ be a basis for $\mathbb{R}^3$. Compute $[T]_{\mathcal{B}}^{\mathcal{C}}$.

$\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$ $\quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 1\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 0\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 0\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 1\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

$[T]_{\beta}^{C} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$

$T(1,1) = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \to \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ $\quad T(2,5) = \begin{pmatrix} 9 \\ 5 \\ -3 \end{pmatrix} \to \begin{pmatrix} 2 \\ 7 \\ 5 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ $\quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \\ 5 \end{pmatrix}$

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4. (20 points) Each part is worth 10 points. Let $T : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ be defined as $T(ax^2+bx+c) = -cx^2+bx-a$.
   (a) Find a basis $\mathcal{B}$ for $P_2(\mathbb{R})$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix.
   (b) Compute $[T^k]_{\mathcal{C}}^{\mathcal{C}}$ where $\mathcal{C}$ is the standard ordered basis $\mathcal{C} = (1, x, x^2)$. Note that you will have different answers for odd k and even k.

a) $std = \{1, x, x^2\}$

$T(1) = -x^2$
$T(x) = x$
$T(x^2) = -1$

$$[T]_{std}^{std} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

$$\det([T]_{std}^{std} - \lambda I) = \begin{vmatrix} -\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ -1 & 0 & -\lambda \end{vmatrix} = \lambda^2(1-\lambda) - (1)(1-\lambda)$$

$$= (1-\lambda)(\lambda^2-1) = 0 \qquad \lambda = 1, -1$$

$\lambda = 1$

$$\begin{pmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & -1 \end{pmatrix} v = 0 \qquad v = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$T(-x^2+1) = -x^2+1$
$T(x) = x$
$T(x^2+1) = -x^2-1$

$\lambda = -1$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix} v = 0 \qquad v = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\beta = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\} \to \beta = (-x^2+1, x, x^2+1)$$

b) $D = Q^{-1} A Q$

$$[T^k]_C^C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}^k \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1}$$

$[T^k]_C^C = ([T]_C^C)^k$

$[T]_B^B = \cancel{Q^{-1}AQ}\ Q^{-1}[T]_C^C Q$

~~turn back~~
(on scratch)

$[T]_C^C = Q[T]_B^B Q^{-1}$

$([T]_C^C)^k = (Q[T]_B^B Q^{-1})^k = Q([T]_B^B)^k Q^{-1}$

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5. (10 points) Let $T : V \to V$ be a linear operator on a vector space V. Let $v \in \ker(T)$ be non-zero and w any vector not in $\ker(T)$. Prove that $\{v, w\}$ is linearly independent.

Let $v \in \ker(T)$ and $v \neq 0$ for $T \in \mathcal{L}(V)$.

Then, $T(v) = 0$.

contradiction proof

Let $w \notin \ker(T)$. This means $T(w) \neq 0$.

Now, assume by contradiction that $\{v, w\}$ is not linearly independent.

Then $av + bw = 0$ for some non-zero a, b.

This means $av = -bw$

$v = -\frac{b}{a} w$ so v is some scaling of w.

Let $\alpha = -\frac{b}{a}$. Then $v = \alpha w$.

Since $T(v) = 0$, $T(\alpha w) = 0$ so

$\alpha T(w) = 0$ and $T(w) = 0$.

This means $w \in \ker(T)$ $\rightarrow\leftarrow$

Thus, since there is a contradiction, $\{v, w\}$ is linearly independent.

Non contradiction proof:

We want to show $av + bw = 0$ only when $a, b = 0$.

Let $T(av + bw) = 0$ for some a, b.

Then $T(av) + T(bw) = 0$

$aT(v) + bT(w) = 0$

Since $T(v) = 0$, $bT(w) = 0$ so $b = 0$ since $T(w) \neq 0$.

Since $b = 0$, if $av + bw = 0$, then $av = 0$. Since $v \neq 0$, then $a = 0$ as well. Since ~~we~~ we have shown that $a, b = 0$, $\{v, w\}$ are linearly independent

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6. (20 points) Each part is worth 10 points. Let W be the subspace of $M_{2\times 2}(\mathbb{R})$ with basis

$$\mathcal{B} = \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \right).$$

Endow the vector space $M_{2\times 2}(\mathbb{R})$ with the inner product $\langle A, B \rangle = \operatorname{tr}(B^t A)$ where tr is the trace or the sum of the diagonal entries of a matrix.

(a) Find an orthonormal basis for W.

(b) Find the vector in W that is closest to the vector $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$ with respect to the inner product. You do not need to prove that the vector you find is the closest.

a) $\beta = \left(\underset{v_1}{\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}}, \underset{v_2}{\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}} \right)$

$\|v_1\|^2 = \langle v_1, v_1 \rangle = tr(v_1^t v_1) = tr\left(\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right) = tr\left(\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right) = 3$

$u_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$w_2 = v_2 - \langle v_2, u_1 \rangle u_1 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} - \left\langle \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} - \frac{1}{3} tr\left(\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \right) \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} - \frac{1}{3} tr\left(\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \right) \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$

$\|w_2\|^2 = tr\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \right) = tr\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = 3$

$u_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$

orthonormal basis $= \left(\frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \right)$

b) $\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle = \frac{1}{\sqrt{3}} tr\left(\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \frac{1}{\sqrt{3}} tr\left(\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \right) = \frac{2}{\sqrt{3}}$

$\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \right\rangle = \frac{1}{\sqrt{3}} tr\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \frac{1}{\sqrt{3}} tr\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \right) = 0$

$\left(\frac{2}{\sqrt{3}}, 0 \right) \to \frac{2}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \frac{2}{3} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

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4b)

$$[T^k]_c^c = \begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^k\begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}^{-1}$$

$$\begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}\begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}^{-1} = I$$

$$\begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}\begin{pmatrix}\frac{1}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{1}{2}\end{pmatrix} = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

$$\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^1 = \begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$$

$$\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^2 = \begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix} = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} = I$$

$$\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^3 = \begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix} I = \begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$$

For odd k: $\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^k = \begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$

For even k: $\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}^k = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$

odd k:

$$[T^k]_c^c = \begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}\frac{1}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{1}{2}\end{pmatrix} = \begin{pmatrix}1&0&-1\\0&1&0\\-1&0&-1\end{pmatrix}\begin{pmatrix}\frac{1}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{1}{2}\end{pmatrix}$$

$$= \begin{pmatrix}0&0&-1\\0&1&0\\-1&0&0\end{pmatrix}$$

even k:

$$[T^k]_c^c = \begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix} I \begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}^{-1} = I = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$